

Baltimore Ravens Data Scientist Application Project

Technical Report

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The Problem:

The ability of a Cornerback to blanket a Wide Receiver in Man-to-Man coverage, and prevent targets entirely is an incredibly useful skill. However, evaluating this skill is difficult, because traditional metrics do not account for the quality of receiver faced, or contextual information such as the route run. This project aims to separate and evaluate **the ability of the cornerback to prevent targets in man-to-man coverage**, accounting for influences outside their control. The analysis does not account for any coverage ability after the ball is in the air, such as deflecting passes or intercepting a pass at the catch point.

Methods:

Dataset Generation:

The first step was to filter and merge the datasets properly to access the information relevant to my project. The data was subsetted to only look at plays classified as Man-coverage for the overall defense, and then subset the player plays dataset to isolate CBs playing Man-coverage on WRs only. WR info was merged into the dataset by means of the `pff_primaryDefensiveCoverageMatchupNflId` field, used to identify who the CB was covering on the given play. Many of the contextual features built into the dataset came from the player plays dataset: the route ran, time to throw, drop back type, and play action indicator. Other features were engineered from the tracking data to isolate WR skill and tendencies.

While features such as yards per route run, targets per route run, and average depth of target can be highly predictive and indicative of a WR's talent level and role within the offense, this data is often not publicly available. As such, I derived values for each WR based on the tracking data provided by the big data bowl. To prevent data leakage and inflated accuracy scores, I used rolling averages of each metric, only accounting for routes completed prior to the given play. As such, many of the estimates are based off very small sample sizes, and thus very noisy. To account for this noisiness, I used a shrinkage factor to pull all estimates towards the league mean values, weighted by the number of routes run. As the sample size for a player increases, the shrinkage becomes less prevalent and their estimate stabilizes. Finally, I added a tracking metric for the depth of the WR at the final moment before a potential target to account for the difficulty and decreased likelihood of deep throws compared to safe underneath throws. It should be noted that I defined the time of the potential target as the moment the ball was thrown to any player, or the opportunity to throw was ended (via a sack, QB run, out of bounds, etc).

The binary response variable was defined as whether the primary matchup WR was targeted on play, 1 for a target and 0 for no target. Plays with no throw (sacks, QB scrambles, etc) were also included as 0's, as theoretically the CB was able to cover their matchup well enough to prevent a throw up until that point. To ensure proper data analysis, the final dataset was split into a training set, validation set, and testing set. The model was trained only on the training set, while the validation set was used for deciding optimal classification cutoffs, and the test set was used solely for reporting accuracy on unseen data.

Model Architecture:

The modeling framework is GPBoost, which combines gradient boosted decision trees with Gaussian process random effects in a unified likelihood framework. The outcome variable is binary: whether the primary coverage matchup receiver was targeted on a given play, modeled with a Bernoulli likelihood and logit link function. The features engineered above were included in the model as fixed effects, accounting for many of the factors outside the CB’s control. Crossed random effects are specified for cornerback id and receiver id, allowing the model to jointly estimate player-level deviations from the population mean after accounting for fixed effect covariates. Cornerback and receiver effects are separately identified through the diversity of matchups in the data — cornerbacks in the sample faced an average of 13 unique receivers, and receivers faced an average of 12 unique cornerbacks, providing sufficient cross-matchup variation for effect separation. Finally, cornerback random effects are extracted on the log odds scale and standardized to a relatively 0-100 skill score using a z-score transformation centered at 50 with a scaling factor of 15, similar to standardized scoring systems such as IQ. Percentile ranks are also reported relative to all cornerbacks in the sample.

Results:

The model is well calibrated on the test set, accurately describing target probability buckets from 10% to 50% (Figure 1). This is the most important metric for this model: generalizing probabilities of targets well to unseen data.

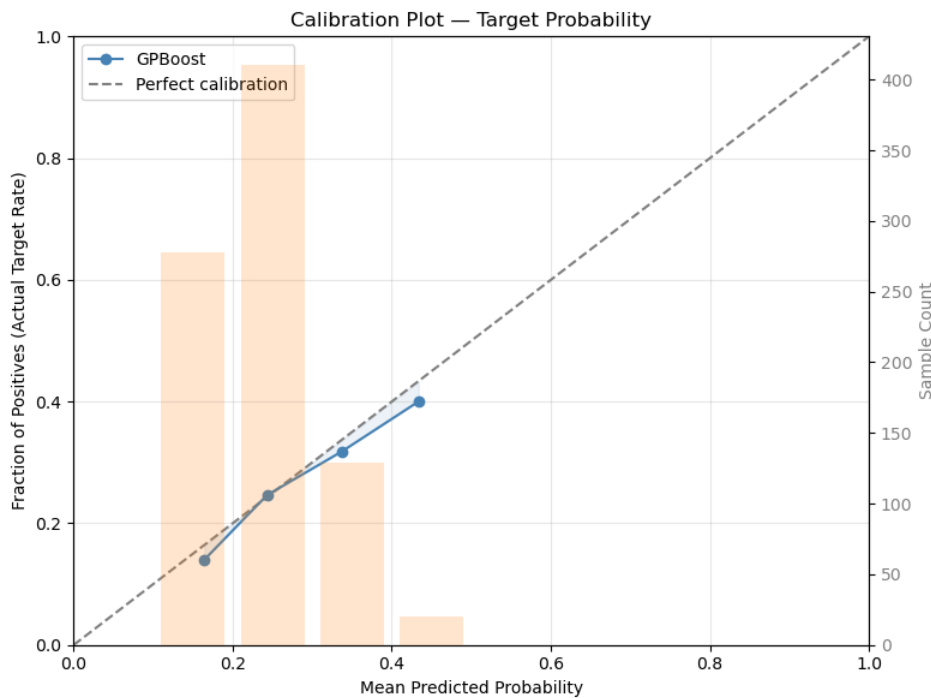


Figure 1

Classification accuracy was moderate, although this is to be expected given the nature of the model design. Because the design was set up to evaluate cornerback skill by absorbing residual

variance into the random effects of the cornerback ids, no tracking metrics for the CB were included, and very few for the WR were included. Including additional tracking information regarding player locations and velocities may improve classification accuracy, but would muddy any interpretation of the resulting talent evaluations. That being said, the model shows real trends in identifying players more or less likely to be targeted, as seen in the predicted probability distribution chart, separated by targeted and non-targeted players (Figure 3). The brier score of the model on unseen data was 0.1691, lower than the expected brier score if every target were treated equally (0.1771), indicating meaningful signal in target probability prediction. Due to the heavy class imbalance between targeted and non-targeted routes (approximately 77% of routes are not targeted), identifying the correct classification threshold was necessary for discrete analysis (Figure 2). The threshold was chosen based on maximum F1 score, using the validation data set, balancing both precision and recall.

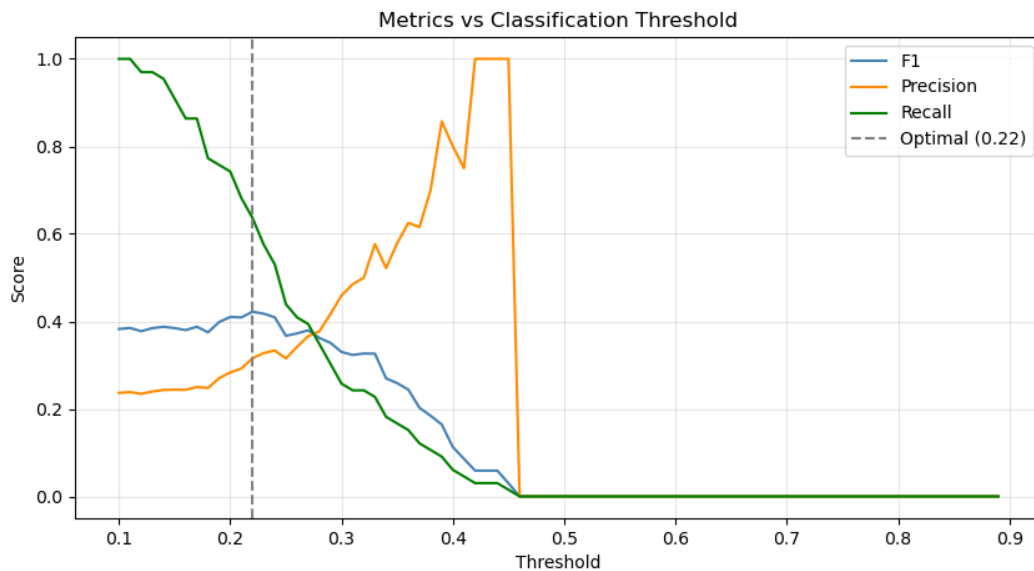


Figure 2

The reduced classification threshold allows the model to correctly identify the majority of targeted routes in the test set (68% recall), but at the cost of precision: many plays predicted as targeted are not actually targeted (29% precision), reflecting the difficulty of identifying the minority class in an imbalanced dataset.

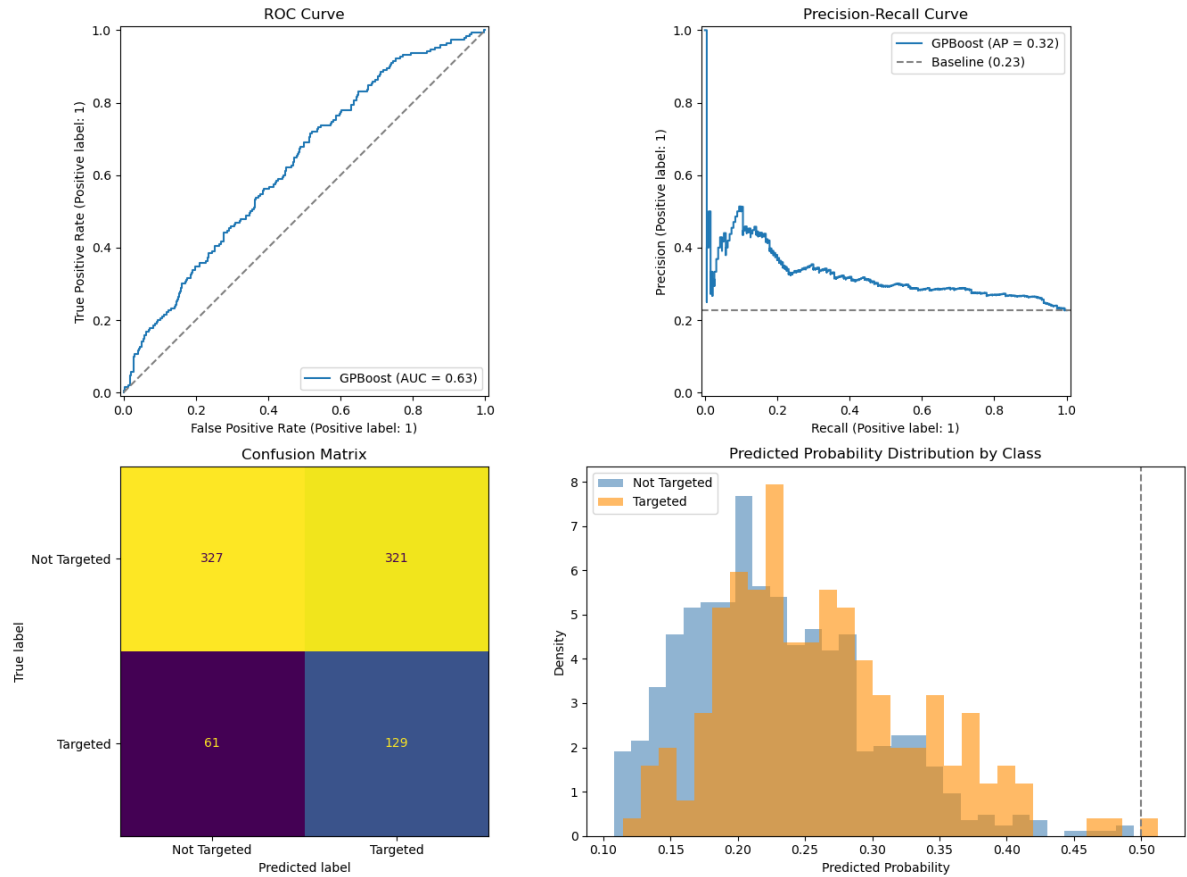


Figure 3

With calibration and discrimination validated, the cornerback random effects can be interpreted with confidence. The z-score transformed random effects and corresponding percentiles, ranking the ability of these cornerbacks to reduce target probability in man-to-man coverage, can be seen below. Players with less than 30 routes defended are excluded to reduce small sample noise.

Top 10 CBs in Preventing Targets in Man-Coverage

Name	Skill Score	Percentile	# Routes
Marlon Humphrey	90.02	100	86
D.J. Reed	89.19	99.4	73
Jonathan Jones	87.63	98.9	89
Arthur Maulet	81.02	98.3	65
Adoree' Jackson	80.44	97.7	121
Mike Hilton	79.97	97.2	65
Mike Hughes	78.38	96.6	62
Darnay Holmes	77.33	96	80
Darius Slay	77.08	95.5	55
Eli Apple	76.94	94.9	70

Validating these results can be difficult, because we are measuring a subskill of CBs that is not typically ranked. However, we can use tangentially related metrics such as PFF grades to get a sense of the accuracy of the model rankings. Three of the top five scorers in the model (Humphrey, Reed, Jackson) scored in the top 21 of PFFs final 2022 CB rankings (<https://www.pff.com/news/nfl-final-2022-cornerback-rankings>), joined by Darius Slay from our top 10. Additionally, top ranked players according to PFF, such as Sauce Gardner and Patrick Surtain II, ranked in the 77th and 66th percentiles respectively in my model. With a larger sample of games and routes, I would expect these rankings to stabilize further, correlating with traditional metrics such as PFF grades while potentially uncovering new insights as well.

Limitations:

The biggest limitations of this model are the small sample sizes used to derive the model estimates. Because the dataset consists of only 9 weeks of data, and only man-coverage plays, the vast majority of corners in this dataset have less than 100 routes defended. Additionally, many of these routes are against the same WRs, limiting the generalizability of the CB estimates. Utilizing a full season of data, or even multiple seasons, would allow for significantly more routes, and more unique CB-WR matchup pairs to strengthen estimates.

Additionally, by design, this model only evaluates man-coverage ability, specifically the ability to prevent a WR from seeing targets. As such, it does not capture any of a player's abilities in zone coverage, or their ability to make plays once the ball is in the air. If desired, these skills would need to be estimated separately to get an idea of the full portfolio of skills each CB possesses.